# Chansoo (Charles) Kim

Graduate Researcher, Liquid-Metal Electronic Materials

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## SUMMARY

Graduate researcher at Yonsei University (BLISS Lab, Prof. Jungmok Seo) developing sintering-free liquid-metal inks, wearable bioelectronics, and miniaturized electrochemical sensor toolkits. Bridging materials chemistry, embedded firmware, and biointerface engineering.

## EDUCATION

2025 – present

**M.S. candidate, Electrical & Electronic Engineering (Liquid-Metal Electronic Materials track)**

Yonsei University — BLISS Lab · Seoul, Republic of Korea

- M.S. thesis track; degree expected February 2028
- Advisor: Prof. Jungmok Seo (BLISS Lab)
- Research focus: sintering-free liquid-metal inks, wearable bioelectronics, electrochemical biosensing

2019 – 2025

**B.S., Electrical & Electronic Engineering**

Yonsei University · Seoul, Republic of Korea

- Bachelor of Science, College of Engineering

## EXPERIENCE

2024-09 – 2025-12

**Undergraduate Laboratory Instructor (Teaching Assistant) — Bio-electrical Electronics Lab**

Yonsei University · School of Electrical & Electronic Engineering · Seoul, Republic of Korea

Led weekly laboratory sessions for the *Bio-electrical Electronics Laboratory* (바이오전기전자실험) course, translating research-grade biopotential / electrochemical / embedded instrumentation into hands-on exercises for undergraduate teams. Designed lab handouts, debugged student hardware and firmware, and mentored end-of-semester project work.

2020-04 – 2021-10

**Training Operations Specialist — Education & Training Command (행정병)**

Republic of Korea Army · 35th Infantry Division Headquarters · Republic of Korea

Active-duty military service supporting divisional education and training operations at the Headquarters' Education & Training Command. Coordinated training schedules, instructor assignments, and curriculum logistics across the division.

2020-08 – 2021-07

**English Interpreter — U.S.–ROK Combined Joint Training Exercise**

U.S. Forces Korea ↔ Republic of Korea Army · Republic of Korea

Selected as combined-exercise interpreter for two annual cycles (Aug 2020 and Jul 2021, ~1 month each), bridging communication between U.S. and ROK forces during joint operations.

2019-05 – 2019-12

**Cultural Programs Coordinator — Cultural Planning Division (문화기획팀)**

Yonsei University · College of Engineering Student Council "PAGE" · Seoul, Republic of Korea

Programmed and produced cultural events for the engineering student community as part of PAGE's freshman-year cultural planning division.

## AWARDS & CERTIFICATIONS

2026-06

**Outstanding Research Team Award — Capstone Design (전기전자종합설계)**

School of Electrical & Electronic Engineering, Yonsei University · Seoul RISE-Y Program

Selected as one of eight outstanding capstone research teams, serving as the team's supervising teaching assistant. The award carries a teaching-assistant scholarship administered through the Seoul RISE-Y program.

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| 2023-08 | **PTRD Scuba Diving Instructor Certification**<br>PTRD (2nd cohort, Scholarship Instructor Program)<br>Earned the *Instructor* rating through PTRD's selective Scholarship Instructor Program (2nd cohort). |

## SERVICE & LEADERSHIP

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| 2023-03 to 2023-08 | **President — Yonsei Skin-Scuba Diving Team (연세스킨스쿠버다이빙팀)**<br>Yonsei University Central Club<br>Headed one of Yonsei's central student clubs; organized training dives, member certifications, and outreach programs for the campus diving community. |

## RESEARCH AREAS

### Liquid-Metal Materials

Catechol-$Ga^{3+}$ chelation chemistry for sintering-free, primer-free EGaIn inks (PDA-LM). DFT (ORCA r2SCAN-3c) studies of amorphous $Ga_2O_3$ surface adhesion.

### Wearable Bioelectronics

Stretchable / tattoo-sticker form factors integrating AD5941 (EC), ADS131M08 (EEG/EMG), and RHS2116 (neurostim). Apollo510 / DA14531 BLE.

### Electrochemical Biosensors

AD5941-based 4-channel portable potentiostat (CV / amperometry / EIS / OCP). DIY SEBS-based screen-printed electrodes; $Na^+$ ion-selective sensing.

### Biointerface Coatings

PDRN + lubricant dual coatings for anti-fouling and tissue regeneration on suture and intramuscular EMG electrodes.

## TECHNICAL SKILLS

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| **Materials & Chemistry** | EGaIn / liquid metal processing, Catechol-metal coordination, PDA / DOPA chemistry, Screen-printed electrodes (SPE), Lubricant / anti-fouling coatings |
| **Computational** | DFT (ORCA r2SCAN-3c, wB97X-D3/TZVP, BSSE), Python (data analysis, plotters), MATLAB |
| **Embedded & Hardware** | AD5941 (CV/CA/EIS/OCP), ESP32-C6, Apollo510, ADS131M08, RHS2116, KiCad PCB / sPCB / FPCB design, BLE (nRF52 / DA14531) |
| **Characterization** | UV-Vis, ITC, XPS interpretation, EIS, Cyclic voltammetry |

## LANGUAGES

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|---|---|
| **Korean** | Native |
| **English** | Professional working proficiency (served as interpreter during U.S.–ROK Combined Exercises) |